

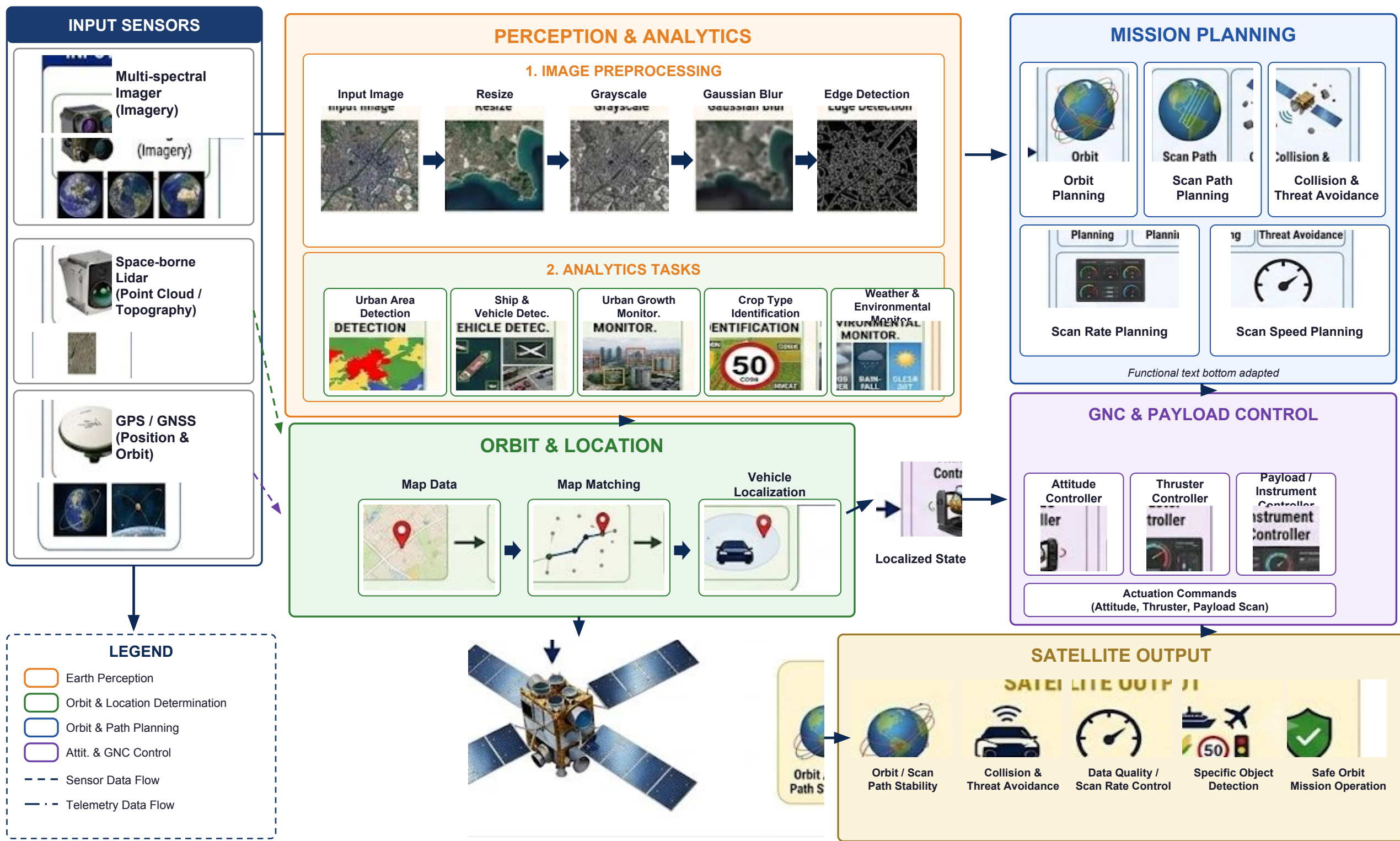
**AT2 (CO1, 20%): Conceptual
Architecture Mapping of Domain
Based Terrain Mapping**

Name: K Balaji

Reg.no:192324180

Course code:DSA0203

SATELLITE-BASED REMOTE SENSING SYSTEM ARCHITECTURE



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The Satellite-Based Remote Sensing System Architecture is designed to collect, process, and analyze Earth observation data for various applications. The system begins with input sensors such as multispectral cameras, space-borne LiDAR, and GPS/GNSS, which capture high-resolution images and location information. The collected images undergo preprocessing steps including resizing, grayscale conversion, Gaussian blur, and edge detection to improve image quality and reduce noise. Advanced AI and computer vision algorithms then perform analytics tasks such as urban area detection, vehicle and ship identification, crop type classification, urban growth monitoring, and weather analysis.

The processed information is integrated with orbit and location data for accurate positioning and mapping. The mission planning module optimizes orbit paths, scan rates, and collision avoidance strategies. Finally, the Guidance, Navigation, and Control (GNC) system controls satellite attitude, payload operations, and thrusters. The system delivers accurate mapping, environmental monitoring, disaster management support, and efficient satellite mission operations.